GenAI for Engineers Learner Guide

IES | v3.3 | Clear explanations and worked examples

One day, 09:00–17:00. Follow the supplied six-module lesson plan. Guided Excel exercises total 65 minutes: SPC 25, FMEA 20, DMAIC 20. The Copilot Agent is a guided paired build in Module 4; SharePoint is a guided 45-minute knowledge-base and citation activity in Module 5.

How to use this guide

The numbered topics match the presentation. First understand the question, then read the example and check the result. Use the separate Starter.xlsx in each lab. Every workbook has only the source table, one prompt sheet, and the pages needed for that activity. Starter.xlsx is optional after-class practice.

Module 1

1. GenAI for Engineers

Use Copilot to understand data and explain engineering findings.

Today: One pump case, three short Excel exercises

Your role: Check the answer before using it

Explanation: All pump data are synthetic.

2. What will we do today?

Follow the six modules in the supplied lesson plan.

Morning: Meet Copilot → write useful prompts → analyse the pump case

Afternoon: Watch agent and SharePoint examples → review AI security

Hands-on work: SPC 25 min · FMEA 20 min · DMAIC 20 min

Explanation: Here, demonstrations and discussion make up the rest of the day.

3. What is generative AI?

It produces a draft answer from your instructions and available information.

You provide: A question and relevant information

AI produces: A draft explanation, table, formula or report

You check: Does the draft agree with the information?

Explanation: A prompt is the instruction you give the AI. Copilot is Microsoft's AI assistant. A fluent answer can still contain a wrong number or unsupported statement.

4. What can Copilot do in Excel?

For example: help you understand why a pump test result looks unusual.

Ask: "Which test results are below the usual range?"

Copilot helps: Suggest a formula or explain the chart

Engineer checks: Compare the answer with the recorded test values

Explanation: These are steps in a joint workflow, not three capabilities belonging to Copilot. Here, the last step belongs to the engineer. Editing and chart features depend on the account and Excel version.

5. Our case: is the pump delivering consistently?

A factory team tests a small pump that feeds an industrial cooling-water system.

The job: The pump should deliver about 62 litres per hour during the test.

The concern: Some later tests record less liquid than earlier tests.

Our task: Describe the change, review possible failures and summarise the evidence.

Explanation: CW-401 is only the fictional case identifier; it is not a method learners must know. A dosing skid is a pump and associated equipment mounted together. We will say pump system in class. These are delivery tests, not measurements of water-treatment compliance.

6. How is the test result measured?

Collect the liquid for two minutes, then convert the amount to litres per hour.

Measured: 2.067 litres collected in 120 seconds

Calculation: 2.067 × 3600 ÷ 120 = 62.01 L/h

Meaning: At that rate, the pump would deliver about 62 litres in one hour.

Explanation: L/h means litres per hour. This is an example calculation, rounded from the synthetic case. A test with a different duration cannot be compared using the same multiplier.

7. Use one prepared workbook today

The calculations and charts are already prepared so we can focus on asking and checking.

Open: A working copy of Worked-Reference.xlsx

Find: Pump_Readings in Lab 1, FMEA_Input in Lab 2, and Weekly_Data in Lab 3.

Then: open the matching lab workbook and select its named table before prompting.

Explanation: Starter.xlsx is optional practice after class.

Module 2

8. A prompt is an instruction to Copilot

A useful prompt tells Copilot what to use and what answer you need.

Too vague: “Analyse this.”

More useful: “Use the pump test table. Identify unusual results and explain them in three sentences.”

Why this helps: Copilot knows the information to use and the size of the answer.

Explanation: Prompt engineering means deliberately writing, checking and improving instructions. It does not mean using complicated language. The improved prompt is still a draft: add the relevant sheet and check a value.

9. Use RISE to build a clear prompt

RISE is a four-part checklist for prompts that produce reviewable engineering work: Role, Input, Steps and Expectations.

R - Role: “Act as an engineering data assistant.” The role sets the viewpoint; it does not give Copilot professional authority.

I - Input: “Use only PumpReadings: eight Baseline and four Monitoring rows.” Name the source, revision, units and boundaries.

S - Steps: “Calculate the baseline mean, sample standard deviation, UCL and LCL; flag monitoring rows; create a line chart.” Put the work in a checkable order.

E - Expectations: “Show formulas, return three sentences, and do not claim a root cause or safety conclusion.” Define the format, quality limit and acceptance check.

Weak prompt: “Analyse this pump data.”

RISE prompt: “Act as an engineering data assistant. Use only PumpReadings. Calculate the baseline mean and control limits, flag the monitoring rows, and create a line chart. Show formulas and give three cautious sentences. Do not claim a root cause.”

Iterate after checking: identify one missing item, correct the instruction, rerun the prompt, and compare the new output with the acceptance check.

10. Add the engineering details with E-MAP

E-MAP reminds you to include context, measures, assumptions and limits.

Engineering + Metrics: “This is a pump delivery test. Report flow in litres per hour.”

Assumptions: “The prepared chart uses the eight Baseline readings as its reference.”

Parameters: “Use the eight Baseline rows to calculate the limits; keep the units in L/h.”

Explanation: E-MAP means Engineering context, Metrics, Assumptions and Parameters. It complements RISE; learners do not have to memorise two competing recipes. A parameter is a value or rule the analysis must use.

11. Tell Copilot what it must not change

Constraints are simple rules that keep the task within scope.

Protect the data: “Do not change the original test records.”

Handle gaps: “If a measurement is missing, say it is missing; do not invent it.”

Limit the conclusion: “Describe the change. Do not claim that we have proved its cause.”

Explanation: Replace abstract warnings with the three exact prompt sentences. Here, a blank measurement means we do not know its value. Removing an extreme valid reading would hide a possible problem.

12. Show the format you want

Use three fields: observation, value and follow-up question.

Requested format: Observation | Supporting value | Question to investigate

Observation + value: T09, T10 and T12 are below the 61.4 L/h lower control limit.

Follow-up question: Were those tests performed under the same conditions?

Explanation: This is a format example, not a complete analysis. Use a short output to make verification practical.

13. Use three perspectives to improve a review

A persona is a requested viewpoint, not a separate expert.

Reliability engineer: “What equipment issue could explain low delivery?”

Operations engineer: “Were the test conditions and settings consistent?”

Safety reviewer: “What must be checked before changing the equipment?”

Explanation: Copilot simulates all three perspectives, so their agreement is not independent evidence. Use the same pump information for all roles.

14. Let the perspectives challenge one claim

Example claim: “Replacing the strainer solved the problem.”

Reliability asks: “Do we have evidence that the removed strainer was blocked?”

Operations asks: “A verification check also happened. Which change made the difference?”

Better conclusion: “Results improved afterwards, but we cannot isolate the cause yet.”

Explanation: A strainer catches particles before they enter equipment. In this case strainer replacement and verification happened together. Do not run an unbounded debate or force consensus.

15. Why spend time improving a prompt?

Clear instructions make the answer easier to use and easier to check.

Less rework: Specify the right sheet, units and output length at the start.

More consistency: Reuse the same structure for similar engineering tasks.

Better review: Ask for supporting values and unanswered questions.

Explanation: Measure total time including checking and correcting the answer. Prompt quality does not guarantee factual accuracy.

16. Turn a test note into a short report

Trainer example: preserve the facts while improving the writing.

Source note: “Later tests lower. Strainer replaced. Verification also done. Pilot results improved.”

Prompt: “Write a short observation and one follow-up question. Use only these facts.”

Example draft: “Delivery improved after two changes. Which change contributed to the improvement?”

Explanation: Here, polishing the language does not create new evidence. The same pattern works for inspection notes and technical reports. This is a demonstration, not another learner lab.

Module 3

17. Three methods answer three different questions

We use the same pump case for all three.

SPC: Has the process changed over time?

FMEA: What could go wrong, and what should we review?

DMAIC: How do we organise an improvement investigation?

Explanation: The full names are: Statistical Process Control; Failure Mode and Effects Analysis; Define, Measure, Analyse, Improve, Control. These are engineering methods. Copilot helps with their outputs; it does not replace learning what the methods mean.

18. SPC: Statistical Process Control

Compare measurements over time with the earlier, usual pattern.

Centre line: The average of the earlier reference tests.

Upper and lower limits: The expected range calculated from that reference period.

A point outside: A reason to investigate a change, not proof of a specific fault.

Explanation: Baseline means the earlier period used for comparison. Here it contains 60 valid tests. Common-cause variation as the usual small fluctuations; an unusual pattern may indicate a special cause. A point inside the limits does not prove the equipment is safe.

19. Read the pump chart from left to right

The later monitoring results drop below the earlier pattern.

Earlier tests: Average about 62.00 L/h

Monitoring period: T09, T10 and T12 are below the lower control limit.

Pilot period: Results return closer to the earlier pattern after two changes.

Control limits: 61.4-62.6 L/h, calculated from the eight Baseline readings.

20. A control limit is not a specification

One describes process behaviour; the other describes the required result.

Control limits: 61.4-62.6 L/h, calculated from the eight Baseline readings.

Test specification: 58.9–65.1 L/h, a fictional requirement supplied for this exercise.

Example: A 59.5 L/h result meets the test specification but is unusual for this process.

Explanation: Cp and Cpk compare process spread and centring with specification limits. They only make sense when the process and measurement assumptions are suitable.

21. Exercise 1: explain the pump chart

25 minutes · use the prepared workbook, not a blank analysis.

Ask Copilot: “Act as an engineering data assistant. Use only the named Excel table PumpReadings. Use the eight Baseline rows to calculate the mean, sample standard deviation, UCL = mean + 3 x standard deviation, and LCL = mean - 3 x standard deviation. Flag each Monitoring row as Below LCL, Above UCL, or Within limits. Create a line chart of all 12 readings with the centre line, UCL and LCL. Return a three-sentence interpretation. Do not claim a root cause or that the pump is safe. Show the formulas used.”

Use the one prompt in Copilot_Prompt. Verify T10 = 59.4 L/h against the calculated LCL. Expected result: T09, T10 and T12 are below the LCL. Save one cautious interpretation.

Save: What changed, the supporting evidence and one unanswered question.

Use the one prompt in Copilot_Prompt. Verify T10 = 59.4 L/h against the calculated LCL. Expected result: T09, T10 and T12 are below the LCL. Save one cautious interpretation.

22. FMEA: think through possible failures

FMEA stands for Failure Mode and Effects Analysis.

Failure mode: The pump delivers too little liquid.

Effect: The system receives less liquid than intended.

Possible cause + control: A restriction may contribute; a delivery test may detect the problem.

Explanation: Separate what failed, its consequence, a possible cause and a control. The potential restriction is not proven. Other FMEA rows can describe possible failures that have not occurred. This prevents learners treating every row as an incident record.

23. Use the scores to prioritise a review

RPN is a risk priority number: Severity × Occurrence × Detection.

Severity (S): How serious is the effect?

Occurrence (O) + Detection (D): How likely is the failure? How hard is it to detect before the effect?

Worked example: FM01: 7 × 5 × 4 = 140. The scores are supplied workshop judgements.

Explanation: Higher detection score means harder to detect, not better detection. FM01 is the ID of the low-delivery row. RPN supports prioritisation; it is not a probability or proof of safety.

24. Exercise 2: check two FMEA rows

20 minutes · inspect the supplied table and propose one follow-up.

Ask Copilot: “Act as an FMEA facilitator. Use only the named Excel table FMEAInput and preserve every row and supplied rating. Create an FMEA output table, calculate RPN = Severity x Occurrence x Detection, rank the four rows from highest to lowest RPN, and flag any row with Severity 9 or higher. For each row, propose one evidence request and one practical follow-up action. Do not invent incidents, change ratings, or approve the design. Show the RPN formula.”

Verify: FM01 has RPN 140. FM04 has severity 9 and RPN 90.

Save: One missing piece of evidence and one proposed action for FM01.

Explanation: Consider why the lower RPN row may still need attention: its consequence is more serious. FM04 is a potential leakage scenario, not an observed leak. Leave residual scores unassessed; stop at 12:30.

25. DMAIC: organise an improvement investigation

Move from a defined problem to a plan for maintaining improvement.

Define + Measure: State the problem and measure how often it occurs.

Analyse + Improve: Investigate explanations and propose a change to test.

Control: Decide what to monitor, who checks it and what happens if results worsen.

Explanation: Use the pump example at each stage. Define: delivery-test deviations. Measure: deviations per test opportunity. Analyse: review chart and records. Improve: test a supported proposal. Control: continue monitoring with a named reviewer.

26. Compare rates, not just event counts

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

Before: 93 deviations ÷ 1,008 opportunities = 9.23%

Pilot: 11 deviations ÷ 336 opportunities = 3.27%

Meaning: One logged chance for a test problem. Compare problems ÷ opportunities.

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

27. A Pareto chart shows the largest categories

Start the investigation where most recorded problems occur.

Largest category: Low delivery: 55 of 93 baseline deviations.

Next category: Test validity: 16 of 93 deviations.

Use it for: Choosing an investigation priority, not proving a root cause.

Explanation: Bars show counts, largest first. The line adds their percentages cumulatively. Low delivery and test validity together account for about 76% of the baseline deviations. A category is a description of the problem, not necessarily its cause.

28. Exercise 3: write a short improvement summary

Open Lab-03-DMAIC/Starter.xlsx. Review the six weekly records in Weekly_Data. Use Copilot to calculate phase rates and draft the five-row DMAIC summary.

Ask Copilot: “Act as a Six Sigma project assistant. Use only the named Excel table WeeklyData. Calculate total opportunities, total deviations, and deviation rate for Baseline and Pilot using deviations divided by opportunities. Create a five-row DMAIC summary: Define, Measure, Analyse, Improve, Control. Clearly separate observed facts from hypotheses. State that the short pilot does not prove the cause of the change. Show the formulas and return one next evidence check.”

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

Save: A lower rate was observed; two changes occurred; more evidence is needed.

Explanation: Do not create a full report or new chart.

Module 4

LO4 guided activity - IES Pump FMEA Assistant

Purpose: build and use one Copilot Agent that performs a six-step FMEA-to-report workflow. This is completed within Module 4 and does not add a fourth lab.

1. Open Microsoft 365 Copilot. Under Agents, select New agent. If the tenant routes you to Copilot Studio, continue in the agent designer.

2. Name the agent “IES Pump FMEA Assistant”. Describe its purpose: turn one supplied FMEA row into a checked report draft.

3. Open Trainer-Demonstrations/FMEA-AGENT-INSTRUCTIONS.md and paste the six-step instruction into the Instructions area.

4. Add FMEA-REPORT-TEMPLATE.md as knowledge or copy its headings into the instruction. Use only synthetic training files.

5. Add the starter prompt: “Review FM01 from the supplied FMEA input and prepare a checked draft.” Save the agent.

6. Test FM01. A passing result preserves S=7, O=5 and D=4, calculates RPN=140, keeps the cause as a hypothesis, and waits for APPROVE DRAFT.

7. Test again without the Detection rating. A passing result stops and asks for the missing value. Refine Step 2 if the agent guesses.

8. Reply APPROVE DRAFT only after checking the row and calculation. Confirm that the report uses Status, Evidence, Risk priority, Proposed action, Owner, Due date and Open question.

9. Create, publish or share the agent using the tenant-approved route and approved training users. If publishing is unavailable, retain the saved draft and preview-test evidence and record the limitation.

10. Save evidence: agent name, instruction version, both test results, one correction, deployment status and the final audit summary.

Official guidance

Build your own agent with Microsoft 365 Copilot: https://support.microsoft.com/en-us/Microsoft-365-Copilot/build-your-own-agent-with-microsoft-365-copilot

Microsoft Copilot Studio documentation: https://learn.microsoft.com/microsoft-copilot-studio

Write agent instructions: https://learn.microsoft.com/en-us/microsoft-copilot-studio/authoring-instructions

Publish an agent: https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/publication-publish-agent

LO5 guided activity - Researcher and Analyst

Purpose: use Researcher for a cited technical comparison and Analyst for a checked data analysis. This 20-minute activity reuses the pump case and does not add another lab.

Part A - Researcher, 8 minutes

Open Microsoft 365 Copilot at https://m365.cloud.microsoft/chat. Select Agents, then Researcher. Attach or reference Trainer-Demonstrations/TR-401-Training-Extract.pdf. Paste the Researcher prompt from REASONING-AGENT-PROMPTS.md. Review the one-page claim-source-evidence-limitation table. Open at least one citation and confirm that it supports the adjacent claim. Keep the missing calibration evidence unresolved.

Part B - Analyst, 8 minutes

Under Agents, select Analyst. Attach Labs/Lab-03-DMAIC/Starter.xlsx. Paste the Analyst prompt from REASONING-AGENT-PROMPTS.md. Check that it uses only the six rows in WeeklyData and separates Baseline W1-W4 from Pilot W5-W6.

Verify the result, 4 minutes

Recalculate Baseline = 36/400 = 9.0%; Pilot = 7/200 = 3.5%; change = -5.5 percentage points; relative reduction = 61.1%. The valid conclusion is that the recorded rate is lower in the two pilot weeks. The data do not prove cause or sustained improvement.

Save the Researcher prompt and response, one opened citation, the Analyst prompt and response, the four checks, one limitation and one next evidence check. If either agent is unavailable, record not executed and use the expected checks for discussion.

Official Microsoft guidance for LO5

Get started with Researcher: https://learn.microsoft.com/en-us/microsoft-365/copilot/researcher-agent

Get started with Analyst: https://support.microsoft.com/en-US/Microsoft-365-Copilot/get-started-with-analyst-in-microsoft-365-copilot

Agents built by Microsoft: https://support.microsoft.com/en-us/Microsoft-365-Copilot/agents-built-by-microsoft

29. What is a Copilot agent?

An assistant with reusable instructions and selected information for a recurring task.

Ordinary prompt: You explain the task again each time.

Agent: The task instructions and selected reference material are already prepared.

Example: An FMEA assistant uses an agreed template and rating guidance.

Explanation: This is a trainer demonstration. Instructions are the task rules and knowledge as reference material. An agent still needs testing and human review. Learners build and test the training agent in pairs, then create or publish it only to the approved training audience.

30. Watch a prepared FMEA assistant

Follow one input through to a checked answer.

Input: Pump description, case records and rating guidance.

Agent draft: Failure modes, effects, possible causes and missing information.

Trainer check: Does each statement match the supplied records?

Explanation: Never claim a recorded live run if using a worked example.

33. Research and analysis need different checks

Research asks what a source says; analysis asks what the numbers show.

Research example: “Find the maintenance guidance in the supplied manual and cite the passage.”

Analysis example: “Compare the test rates and show the calculation.”

Check: Open the cited passage; independently calculate one result.

Explanation: The learning point is the difference in evidence, not a guarantee of access. Use prepared source and arithmetic examples if live tools are unavailable.

34. A report assistant can reuse a template

It turns recorded facts into a consistent report structure.

Recorded facts: Observation, action, owner and due date.

Report sections: Progress, outstanding issues and next actions.

Human review: Check that “proposed” has not become “completed”.

Explanation: A minutes-to-actions-to-report workflow with the same fields. Do not imply that recurring automation has been deployed or that the assistant can approve work.

Module 5

LO6 — create and use a SharePoint-grounded engineering knowledge base

By the end of this 45-minute module, you can explain SharePoint grounding, inspect a private knowledge folder, create or review a scoped agent, ask supported and unsupported questions, open a citation, and check revision, scope and permissions.

Why engineers use it

A SharePoint knowledge base keeps approved files in one managed location. A scoped agent can search those files, draft a cited answer and reduce repeated manual searching. The citation gives the engineer a path back to the evidence; it does not approve the conclusion.

Prepared private site

Open https://tertiaryinfotech.sharepoint.com/sites/IES-GenAI-for-Engineers. The prepared route is Documents > Engineering Knowledge Base. Confirm the four synthetic sources: TR-401-Training-Extract.pdf, SPC-Starter.xlsx, FMEA-Starter.xlsx and DMAIC-Starter.xlsx. The prepared agent is IES Engineering Knowledge Assistant. Do not change site membership or permissions during the activity.

Create or inspect the agent

Select the Engineering Knowledge Base folder. Choose AI actions or the folder context menu, then Create an agent. Set the name, purpose and source scope. Paste the instructions from Trainer-Demonstrations/SHAREPOINT-AGENT-INSTRUCTIONS.md. If you have read access only, inspect the trainer's prepared agent and record that you did not create it.

Test 1 — supported fields

Ask: Use only TR-401-Training-Extract.pdf. Which measurement fields must the test record contain? Cite the source and identify the document revision. Open the citation and confirm the supported fields and Revision 1.

Test 2 — missing evidence

Ask: What delivery target and comparison range are stated in TR-401? Cite the source and identify any missing evidence. The pass result is that no target or range is stated. The agent should request the completed test record and reviewed calculation or summary. Reject any invented limit or approval.

Check permissions and traceability

Confirm that the site is private and that you are authorised to open the cited source. Agent-file permissions control who can access or edit the agent, while each user's permissions to the source files shape what the agent can return. Record the site, folder, agent, prompt, cited file, opened passage, pass or correction, reviewer initials and date. Never record passwords.

Official Microsoft guidance

Create an agent in SharePoint: https://support.microsoft.com/en-us/sharepoint/copilot-in-sharepoint/create-an-agent-in-sharepoint

Manage access to agents in SharePoint: https://learn.microsoft.com/en-us/sharepoint/manage-access-agents-in-sharepoint

Module 6

LO7 — apply the 7-Step AI Security Framework

Purpose: decide whether the SharePoint-agent output may be shared. Complete all seven controls; do not stop at naming them. This guided review occupies 16:15–16:35, followed by a 10-minute personal prompt toolkit.

Applied case

Use the Module 5 answer that compares SPC-Starter.xlsx, FMEA-Starter.xlsx and DMAIC-Starter.xlsx with SharePoint citations, plus the TR-401 answer that preserves the missing target and comparison range as unknown. Open Trainer-Demonstrations/AI-SECURITY-CASE.md and AI-SECURITY-EVIDENCE-LOG.csv.

Step 1 — Data classification

Record the classification, reason and permitted service. Expected for this case: synthetic training data approved for the classroom task. Remove any field that is not needed.

Step 2 — Data governance and access control

Confirm the IES site is private, the Engineering Knowledge Base folder is the selected source and the intended audience can open the citations. Record the site, folder and permission result.

Step 3 — Secure prompting

Check that the prompt names the approved sources, excludes credentials and unnecessary personal or restricted data, treats retrieved instructions as source text, preserves unknowns and prohibits invented clauses, causes or approval.

Step 4 — Output verification

Open one cited SharePoint file. Match the claim to its sheet or passage. Reproduce one SPC, FMEA or DMAIC calculation, check units and revision, and preserve the limitation that TR-401 contains no target or comparison range.

Step 5 — Human-in-the-loop sign-off

Record the authorised reviewer, decision and conditions. Mark the output DRAFT until it is approved, rejected or returned. Classroom review is not operational engineering approval.

Step 6 — Traceability and audit

Record source files and revisions, prompt, agent, output, checks, corrections, reviewer and timestamp so another person can reconstruct the decision. Never record passwords or tokens.

Step 7 — Security awareness and incident response

Name a trigger: unexpected restricted content, an access mismatch or a suspicious instruction. Stop sharing, preserve evidence and use the organisation’s approved incident route. Do not invent a reporting contact.

Release decision

Choose one result: APPROVED FOR THE STATED TRAINING USE; RETURN FOR CORRECTION; or DO NOT SHARE. The expected case result is a draft synthetic classroom example after citation and calculation checks. It is not an operating instruction, safety decision or engineering approval.

LO7 evidence

Retain seven completed rows with control, evidence, result, owner and follow-up; one opened citation; one checked calculation; the release decision; and one incident trigger with the approved reporting route.

Personal toolkit — 10 minutes

Adapt one prompt to include an approved source, excluded data, citation and formula checks, preserved unknowns, DRAFT status and an authorised reviewer. Data governance, audit records and incident response remain separate controls.

Course close

Copilot can reduce drafting, research, retrieval and analysis effort. Engineers remain responsible for methods, evidence, decisions, security and release.

Detailed exercise instructions

Lab-01-SPC

SPC — 25-minute guided Excel exercise

Module 3 only. Use the separate Starter.xlsx in each lab; open the Worked-Reference only after the attempt.

5 min · Read the prepared analysis

Open Lab-01-SPC/Starter.xlsx. Review the 12 rows in Pump_Readings: eight Baseline and four Monitoring. Select the named table PumpReadings before opening Copilot.

12 min · Prompt and inspect

Use the one prompt in Copilot_Prompt. Verify T10 = 59.4 L/h against the calculated LCL. Expected result: T09, T10 and T12 are below the LCL. Save one cautious interpretation.

8 min · Verify and retain

Use the one prompt in Copilot_Prompt. Verify T10 = 59.4 L/h against the calculated LCL. Expected result: T09, T10 and T12 are below the LCL. Save one cautious interpretation.

Optional after class — not scheduled

Starter.xlsx and the complete prompts.md bank support rebuilding the analysis, chart and extended review later. Only the selected core prompt is required in class.

Lab-02-FMEA

FMEA — 20-minute guided Excel exercise

Module 3 only. Use the separate Starter.xlsx in each lab; open the Worked-Reference only after the attempt.

4 min · Read the prepared FMEA

Open Lab-02-FMEA/Starter.xlsx. Review the four supplied failure modes and ratings in FMEA_Input. Use Copilot to create the formula-based output; do not invent or rescore rows.

10 min · Prompt and check priorities

Paste the one prompt from Copilot_Prompt. Ask Copilot to calculate RPN, rank the four rows and flag Severity 9 or higher. Independently verify FM01 and FM04.

6 min · Review one row

Write one evidence gap for FM01 and one action proposal. Keep residual ratings unassessed. The multi-persona debate is a trainer demonstration in Module 2, not an extra participant task.

Optional after class — not scheduled

Starter.xlsx and the complete prompts.md bank support rebuilding the analysis, chart and extended review later. Only the selected core prompt is required in class.

Lab-03-DMAIC

DMAIC — 20-minute guided Excel exercise

Module 3 only. Use the separate Starter.xlsx in each lab; open the Worked-Reference only after the attempt.

4 min · Read the prepared rates

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

10 min · Prompt a short summary

Use the one prompt in Copilot_Prompt. Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%. State that the short pilot does not prove the cause of the change.

6 min · State a decision boundary

Save three cells: measured change, limitation and next evidence. The absolute change is 5.5 percentage points. Two concurrent changes and a short pilot do not prove a cause or sustained improvement.

Optional after class — not scheduled

Starter.xlsx and the complete prompts.md bank support rebuilding the analysis, chart and extended review later. Only the selected core prompt is required in class.

Plain-language glossary

Prompt: An instruction or question you give to an AI assistant.

Baseline: The earlier set of results used as the reference for comparison.

Control limit: A boundary calculated from the reference process data to help identify unusual behaviour.

Specification: A stated requirement that the result is expected to meet.

Failure mode: The way a component or process could fail.

RPN: Risk priority number: Severity × Occurrence × Detection.

Grounding: Supplying relevant information, such as a document passage, to help an AI answer a question.

Persona: A requested viewpoint used to review a question, not an independent expert.

Pilot: A limited follow-up or trial period used to collect more evidence.

Worked demonstration examples

31. A reusable instruction for the FMEA assistant

Show the instruction, then test it on just one failure mode.

Use: “Use the case records and supplied rating guidance.”

Return: “Give the failure, effect, possible cause and missing evidence.”

Check: “Keep supplied scores. Do not turn a possible cause into a fact.”

Use Trainer-Demonstrations/README.md for the full prompt. Open FMEA_Input, ScoringNote and Evidence. Test FM01 only. A normal Copilot prompt demonstrates the instruction if an agent is unavailable; identify the surface honestly.

32. What should the FMEA assistant say?

Example question: “What is still unproven about the low-delivery failure?”

Observed fact: Later tests recorded lower delivery.

Possible explanation: A restriction may have contributed.

Still needed: Evidence about the removed strainer and the effects of the two changes.

This is an illustrative response, not a recording of a live agent run. Open FM01 and the maintenance evidence. Ask learners to identify the fact, hypothesis and unanswered question.

35. A report draft that keeps an open issue visible

Use the same three headings every time: status, evidence and next action.

Status: Pilot delivery results improved.

Evidence: Strainer replacement and verification occurred together.

Next action: Complete the removed-strainer review before claiming a cause.

Read the source note in Trainer-Demonstrations/README.md. Show how a report template makes the distinction between observed improvement and proven cause visible. Ask why changing an open review to completed would be an error.

38. Check a grounded answer against a real passage

Use the supplied fictional document TR-401, revision 1.

Question: “Which measurement fields must we record?”

Answer: “Test ID, timestamp, volume in litres and duration in seconds.”

Check: Clause 1 lists those four fields. It does not approve a delivery limit.

Open Trainer-Demonstrations/TR-401-Training-Extract.pdf and read clause 1. Ask the follow-up about an approved limit: the correct answer is not specified. Demonstrate that a citation is useful only when the passage supports the answer.

Using the expanded 257-slide deck

Slides 1–116 are the one-day teaching route. Slide 117 introduces optional reference material. Slides 118–237 contain 30 worked examples; they are not additional scheduled labs. Keep the original six-module timetable and the 25/20/20-minute guided Excel exercises.

Reference index: SPC slides 138-169; FMEA 170-193; DMAIC 194-217; Prompting 218-237; Agents 238-245; Grounding 246-253; Security 254-257.

Each reference example follows data, prompt, illustrative answer and verification. Supporting CSVs, prompts and Worked-Examples.pdf are in Reference-Examples. The trainer demonstrations use the supplied Trainer-Demonstrations pack.

Open Microsoft 365 Copilot

Microsoft 365 Copilot — sign in with your work or school account.

Get started with Copilot in Excel — Excel analysis guidance.

How Copilot Chat works in Microsoft 365 apps — Word, Excel and PowerPoint guidance.

If the Copilot button is missing, use the approved manual-formula fallback and tell the trainer. Availability depends on your organisation’s licence and settings.

Interface screenshots

Use these current Microsoft 365 web interfaces to locate Copilot. Your tenant layout may differ slightly.

Microsoft 365 Copilot home — Chat and Agents are available from the left navigation when enabled.

Excel for the web — the Copilot button is shown at the lower right when available.

LO2 practical workflow: create the engineering output with Copilot 365

Learners repeat the same five actions in all three exercises: open the small Starter workbook; select the named table; open Copilot in Excel; paste the one supplied prompt; inspect and independently verify one result. The Worked-Reference workbook is opened only after the attempt.

Lab 1 - Use Copilot in Excel to create an SPC review (25 minutes)

Source: Pump_Readings / PumpReadings.

Copilot prompt: Act as an engineering data assistant. Use only the named Excel table PumpReadings. Use the eight Baseline rows to calculate the mean, sample standard deviation, UCL = mean + 3 x standard deviation, and LCL = mean - 3 x standard deviation. Flag each Monitoring row as Below LCL, Above UCL, or Within limits. Create a line chart of all 12 readings with the centre line, UCL and LCL. Return a three-sentence interpretation. Do not claim a root cause or that the pump is safe. Show the formulas used.

Independent check: Verify T10 = 59.4 L/h against the calculated LCL. A point below the LCL is a signal to investigate, not a diagnosis.

Lab 2 - Use Copilot in Excel to create an FMEA review (20 minutes)

Source: FMEA_Input / FMEAInput.

Copilot prompt: Act as an FMEA facilitator. Use only the named Excel table FMEAInput and preserve every row and supplied rating. Create an FMEA output table, calculate RPN = Severity x Occurrence x Detection, rank the four rows from highest to lowest RPN, and flag any row with Severity 9 or higher. For each row, propose one evidence request and one practical follow-up action. Do not invent incidents, change ratings, or approve the design. Show the RPN formula.

Independent check: Verify FM01: 7 x 5 x 4 = 140. Also verify that FM04 is flagged because Severity = 9 even though its RPN is lower.

Lab 3 - Use Copilot in Excel to create a DMAIC summary (20 minutes)

Source: Weekly_Data / WeeklyData.

Copilot prompt: Act as a Six Sigma project assistant. Use only the named Excel table WeeklyData. Calculate total opportunities, total deviations, and deviation rate for Baseline and Pilot using deviations divided by opportunities. Create a five-row DMAIC summary: Define, Measure, Analyse, Improve, Control. Clearly separate observed facts from hypotheses. State that the short pilot does not prove the cause of the change. Show the formulas and return one next evidence check.

Independent check: Verify Baseline = 36 / 400 = 9.0% and Pilot = 7 / 200 = 3.5%. The lower pilot rate is an observation, not causal proof.

Microsoft 365 Copilot links

Microsoft 365 Copilot: https://m365.cloud.microsoft/chat

Get started with Copilot in Excel: https://support.microsoft.com/en-us/office/get-started-with-copilot-in-excel

How Copilot Chat works in Microsoft 365 apps: https://support.microsoft.com/en-us/microsoft-365-copilot/how-copilot-chat-works-in-microsoft-365-apps